ROAD BIKES · 6 MIN READ

Groupsets and gearing

The groupset is the shifters, derailleurs, crank, cassette, chain and brakes. It is the biggest reason two bikes with the same frame have different prices.

Shimano road ladder

GROUPSET	GEARS	SHIFTING
Claris	2 × 8	Mechanical
Sora	2 × 9	Mechanical
Tiagra	2 × 10	Mechanical
105	2 × 12	Mechanical or Di2
Ultegra	2 × 12	Di2 only
Dura-Ace	2 × 12	Di2 only

Di2 is Shimano's electronic shifting. **GRX** is Shimano's gravel groupset.

SRAM road ladder

Apex, Rival, Force, Red, from entry to top. **AXS** means wireless electronic shifting. So "Rival AXS" is the Rival level with wireless electronic shifting.

Di2 and AXS are not the same

- Shimano Di2: the shifters are wireless and run on small coin cells, but both derailleurs are wired to one rechargeable battery inside the frame. That one battery is what the rider charges.
- SRAM AXS: fully wireless. Each derailleur has its own clip-on battery, so there are two to charge.

What the customer gets as they move up

More gears with smaller jumps between them, lighter parts, crisper shifting and better brakes. Shimano 105 is where most keen riders stop: it has the same 12 gears as the top groupsets at a much lower price.

Gearing in plain words

- Front chainrings: **50/34** is called compact. **52/36** is semi-compact and suits stronger riders.
- Rear cassette: written as smallest to biggest cog, like **11-34**. A bigger last number means an easier climbing gear.
- The easiest gear is small ring in front, biggest cog at the back. With 34 front and 34 rear the wheel turns once for each pedal turn. That is a good ghat gear.

Remember these

- Claris, Sora, Tiagra, 105, Ultegra, Dura-Ace: low to high.
- 12-speed Ultegra and Dura-Ace are Di2 only. 105 comes mechanical or Di2.
- SRAM: Apex, Rival, Force, Red. AXS means wireless electronic.
- A bigger biggest-cog (like 34) makes climbing easier.